

National University of Computer and Emerging Sciences, Lahore Campus				
	Course:	Machine Learning for Robotics	Course Code:	CS 5096
	Program:	BS (Computer Science)	Semester:	Spring 2026
	Date:	27-04-2026	Total Marks:	10
	Section:	A	Weight:	
	Exam:	Quiz 2	Roll No:	

Q1) Consider a neural network using a unit step activation function. (6)

$$f(z) = \begin{cases} 1 & \text{if } z \geq 0 \\ 0 & \text{if } z < 0 \end{cases}$$

Find a valid combination of weights (w_1, w_2) and bias (b) for each of the following gates. For each gate, show the calculation for the input pair (1,0) to verify your logic. Constraints: The absolute value of w_1 and w_2 must be prime.

a) The OR Gate

b) The NAND Gate (Output is 0 only if both inputs are 1)

Q2) You are designing a neural network for a computer vision task. The network processes a batch of images and passes them through two hidden layers. The input X is a batch of 50 images, where each image is flattened into a vector of 1,024 pixels. Layer 1 (L1) has 256 neurons. Layer 2 (L2) has 128 neurons. Output Layer (L3) is a multiclass classification layer with 10 units (one for each class). The network is implemented using the vectorized formula:

$$Z = W \cdot X + b$$

What are the dimensions of the input matrix X ? (2)

What are the dimensions of the weight matrix ($W[1]$) and bias vector ($b[1]$) for the first layer? (2)